

Vantage Runbook

Complete instructions for standing up a Vantage instance from a fresh machine — with Docker, or fully native with no Docker and no admin rights. Written so someone who has never seen the project can follow it end to end.

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1 What you are running

Vantage is self-hosted and single-tenant: **each person runs their own instance on their own machine, connected to their own account.** Nothing is shared between instances, and sign-in only works from a browser on the machine hosting the app — that is a security feature, not a bug.

Four pieces, identical in both paths:

Piece	What it does
PostgreSQL	Stores everything, encrypted at rest with a key generated at setup
API (FastAPI)	Serves the app and the browser UI on port 8787
Worker	Background sync and daily brief generation
Frontend	Built React app, served as static files by the API

Before either path, you need

- A machine you control (this guide is Windows-centric; macOS/Linux follow the same shape)
- A **Microsoft Entra app registration** (Outlook accounts) or a **Google Cloud OAuth client** (Google accounts) — your own, free, ~10 minutes; details in section 4
- An **LLM API key** — Anthropic (Claude) or OpenAI
- The account to connect. **One instance = one provider = one account**, fixed at first sign-in. Switching later means wiping the instance (Settings → Delete everything)

Choosing a path

	Path A — Docker	Path B — Native
Best for	Personal machines, machines with admin	Corporate laptops, no admin rights
Needs admin?	Usually once (WSL2 / runtime install)	No
Effort	Low	Medium, one-time
After a reboot	Runtime restarts containers	Re-run <code>start</code> manually

2PATH A

Docker

2.1 Install a container runtime (once)

Any of these gives you the same `docker compose` commands:

- **Docker Desktop** — easiest. Free for personal use; needs a paid licence at companies with 250+ employees or \$10M+ revenue.
- **Rancher Desktop** (rancherdesktop.io) — free, no licence restrictions. During first-run setup choose the "**dockerd (moby)**" engine and disable Kubernetes.
- **Podman Desktop** — free; use `podman compose`.

All three need **WSL2** on Windows: `wsl --install` in an admin PowerShell, once, then reboot. If WSL is blocked by IT policy, use Path B instead.

Verify: `docker --version` and `docker compose version` both print versions.

2.2 Run it

From the folder containing `compose.yaml` (the `vantage/` folder):

```
docker compose up --build -d
```

```
docker compose ps
```

`ps` should show `postgres`, `app`, and `worker` running; the `init` container exits after applying database migrations — that is normal. Open **`http://localhost:8787`** and continue at section 4.

2.3 Day-to-day

What	Command
Stop (data kept)	<code>docker compose down</code>
Start again	<code>docker compose up -d</code>
Update after new code	<code>docker compose up --build -d</code>
Logs	<code>docker compose logs -f app worker</code>
Wipe everything	<code>docker compose down -v</code> — deletes the database

Data lives in a named Docker volume; it survives `down/up` and rebuilds, and is destroyed only by `down -v`. A parallel dev instance on port 8788 exists via `docker compose -p vantage-ms -f compose.yaml -f compose.microsoft.yaml up --build -d`.

3^{PATH B} Native — no Docker, no admin

Everything installs per-user. Nothing requires administrator rights.

3.1 Where to put the project — read this first

Put the project at a **short path outside OneDrive**, ideally `C:\Users\<you>\vantage`.
Two hard-won reasons:

1. **Windows' 260-character path limit.** Deep OneDrive paths plus Python package paths exceed it, and `pip` fails with `[Errno 2] No such file or directory`. Raising the limit needs admin; a short path doesn't.
2. **OneDrive syncs live database files.** The app's PostgreSQL data lives inside the project folder; syncing it can corrupt the database and uploads the app's cached mail to corporate OneDrive.

Note

If the project truly cannot be moved out of OneDrive, use the [subst workaround in 3.6](#).

3.2 Install Python 3.13+ (no admin)

1. Download the Windows 64-bit installer from python.org/downloads
2. In the installer: **uncheck** "Use admin privileges when installing py.exe", **check** "Add python.exe to PATH", then **Install Now** — it installs per-user
3. Open a **new** terminal and verify: `py --version` → `Python 3.13.x`

3.3 Install Node.js 20+ (no admin)

The regular `.msi` installer needs admin — use the zip instead:

1. From nodejs.org/en/download choose the LTS "**Standalone Binary (.zip)**" for Windows x64
2. Extract it; note the folder that directly contains `node.exe` (e.g. `C:\Users\<you>\AppData\Local\Programs\node`)
3. Add that folder to your **user** PATH: Start → "Edit environment variables for your account" → Path → Edit → New → paste the folder path
4. Open a **new** terminal and verify: `node --version` and `npm --version`

3.4 Run setup

From the project root — the folder containing `scripts\`, `backend\`, `frontend\` (run `dir` to confirm):

```
powershell -ExecutionPolicy Bypass -File scripts\run-native.ps1 setup
```

Setup, in order: downloads portable PostgreSQL binaries (~350 MB, no installer, no admin), creates the Python environment, installs backend dependencies, builds the frontend, generates the instance secrets, and initialises the database. Ends with `setup complete`. Takes ~10–15 min with the download, ~3–5 min without. Safe to re-run at any time — it resumes where it left off.

```
If the Postgres download fails
```

Corporate proxies often break it. Download the **Windows x86-64** binaries zip manually in a browser from enterprisedb.com/download-postgresql-binaries, right-click → Extract All, and place the extracted `pgsql` folder inside `.native` so that `<project>\.native\pgsql\bin\initdb.exe` exists. Then re-run setup — it skips the download.

3.5 Back up the secrets, then start

Do this once, seriously

Copy `.native\secrets.env` somewhere safe (password manager, personal drive). It holds the encryption key — losing it makes all stored data unrecoverable.

```
powershell -ExecutionPolicy Bypass -File scripts\run-native.ps1 start
```

Wait for running - open `http://localhost:8787`, open that URL, and continue at section 4.

What	Command (from the project root)
Start — after every reboot; nothing auto-starts	<code>powershell -ExecutionPolicy Bypass -File scripts\run-native.ps1 start</code>
Stop	<code>...run-native.ps1 stop</code>
What's running?	<code>...run-native.ps1 status</code>

Logs

`.native\logs\` — `app.err.log`, `worker.err.log`,
`postgres.log`

Everything mutable lives in `.native\` (git-ignored): `pgsql\` (Postgres binaries), `pgdata\` (the database), `secrets.env`, `logs\`, `pids.json`. Deleting `.native` is a factory reset. Never copy one machine's `.native` to another machine.

3.6 Stuck inside a long OneDrive path? The `subst` workaround

If the project cannot be moved and pip fails with the long-path error, map a short virtual drive onto the project folder. In **Command Prompt**, `cd` into the project root, then:

```
subst V: "%cd%"
V:
rmdir /s /q V:\backend\.venv      :: clears any half-built env; "path not found"
is fine
```

```
powershell -ExecutionPolicy Bypass -File scripts\run-native.ps1 setup
```

`v:` is an alias for the same folder — nothing is copied. Every path now starts with `v:\`, which fits under the limit. Python may print "Actual environment location may have moved due to redirects, links or junctions" — informational, not an error.

Rules that come with `subst`

The mapping disappears on reboot — re-run `subst V: "<project path>"` and `V: /start` each time (save the lines in a `.cmd` file). And pause OneDrive sync while the app runs (tray icon → Pause), because the database still physically lives inside OneDrive on this arrangement.

4 First run — the setup wizard (both paths)

The wizard at **`http://localhost:8787`** walks through everything below. Deeper reference lives in the repo at `docs/SETUP.md`.

Microsoft (Outlook) track

1. **Register an Entra application** (once, free): Entra admin center → App registrations → New registration. Supported account types: your organisation only (work) *or* include personal Microsoft accounts (outlook.com). Under **Authentication** → **Add a platform** → **Web**, add the redirect URI exactly, matching your port:
`http://localhost:8787/api/v1/auth/microsoft/callback`
Then **Certificates & secrets** → **New client secret** — copy the *Value* immediately; it is shown once.
2. **Wizard step 1**: paste the client ID and secret. **Tenant field**: leave blank for personal accounts. For a single-tenant work app, enter the Directory (tenant) ID — a single-tenant app used via the default `common` endpoint fails with `AADSTS50194`.
3. **Connect** — sign in. Read scopes requested: `User.Read`, `Mail.Read`, `Calendars.Read`, `Tasks.Read` plus `identity/offline`. **Work tenants may show "need admin approval"** — an admin must consent once; send them the app name, the read scopes, and, if action features are wanted, the write scopes:
`Mail.ReadWrite`, `Calendars.ReadWrite`, `Tasks.ReadWrite`.

4. **Sources:** tick the mail folders, calendars, and To Do lists the app may read — nothing unticked is ever fetched — and **save**.
5. **Intelligence:** choose the LLM provider and paste its API key. Set the account's tokens-per-minute limit here to avoid rate-limit failures on the first brief.

Google (Gmail) track

Same shape: create a Google Cloud OAuth client (see `docs/SETUP.md` Parts 1–2), paste credentials, connect, tick sources, add the LLM key. Redirect URI:
`http://localhost:8787/api/v1/auth/google/callback`

After the wizard

Sync, then build the brief. Grant write permissions individually from **Settings** → **Write permissions** — each is a separate OAuth consent, and every write still requires approving the exact payload in Approval Center before anything touches your account.

5 Troubleshooting

Symptom	Cause and fix
Setup: Postgres download fails / <code>Cannot find path ... clusterdb.exe</code>	Proxy broke the download or antivirus interfered. Download the binaries zip manually in a browser and place <code>pgsql</code> under <code>.native</code> (§3.4)

pip: [Errno 2] No such file or directory + hint about Long Path support	Path too long (OneDrive nesting). Move the project to a short path, or use <code>subst</code> (§3.6). Either way, delete <code>backend\.venv</code> and re-run setup
'Remove-Item' is not recognized	PowerShell command typed into Command Prompt. cmd equivalent: <code>rmdir /s /q <folder></code>
pip/npm TLS or certificate errors	Corporate proxy re-signing traffic — capture the exact error and escalate to IT; usually fixed by trusting the corporate CA
Browser can't reach the app but status says API ok	Try <code>http://127.0.0.1:8787</code> — some tools resolve <code>localhost</code> to IPv6 first
Sign-in ends on AADSTS50194	Single-tenant Entra app used with tenant <code>common</code> — put the Directory (tenant) ID in the wizard's Tenant field
Sign-in ends on "need admin approval"	Work tenant requires admin consent once — see section 4
"Sign in from the machine running Vantage"	By design — sign-in only works in a browser on the machine hosting the app
Sync succeeds but stores 0 items	No sources ticked in Settings — tick folders/calendars/lists and save
Port 8787 already in use	Another instance is running (Docker and native both default to 8787) — stop one
Native: app gone after reboot	Nothing auto-starts — re-run <code>start</code> (and re-run <code>subst</code> first if using §3.6)

6 Data, security, backup

- **Everything stays local** except source excerpts sent to the LLM provider you configured. Mail and calendar content is encrypted at rest; OAuth tokens and API keys are encrypted; account addresses are stored only as HMAC fingerprints.
- **Back up one thing.** Docker: the data volume. Native: `.native\secrets.env` (and optionally `.native\pgdata` while the app is stopped). Losing the encryption key means losing the instance's data.
- **Every write is approval-gated.** Proposals show the exact payload, expire in 15 minutes, and re-check the provider's version before executing. Vantage can create mail drafts but never sends mail without the separate send grant (Google) — and never requests `Mail.Send` at all (Microsoft).
- **Reset or hand-off.** Settings → Delete everything (or `down -v / delete .native`) wipes tokens, cached data, and keys, and returns the instance to unclaimed. Your data at Google or Microsoft is never touched.